SE Senior Project Spring 2026
Tiger Flood Rescue, BAE Systems

Final Plan

Project Goals and Scope Statement

The goal of this project is to create a drone system which is able to detect and report lost or abandoned dogs. This information will be displayed in an Augmented Reality sand table view. The core of this project will be on the data processing and sand table portion, while leveraging pre-existing software for connecting and flying drones.

Initial Project Tools and Techniques

Project Management Tools: Jira - Story tracking, Google Sheets - Time Tracking spreadsheet, PowerPoint - Monthly Reports

Technical Development Tools: GitLab, VScode, IntelliJ, Development Laptops

Technical Implementation Tools: Drones, AR Headsets, Android Phones, Computer Vision Model, Software Framework

Development Methodology

Feature Driven Development

Planned Activities and Artifacts

Activities

1. Feature/Backlog Refinement - Biweekly
2. Feature Planning - Once per feature, more if needed
3. Feature Design Meeting - Twice per feature, more if needed
4. Standup Meeting - Every Tuesday, ~20 minutes

Artifacts

1. Overall system model - Done once
2. Feature List - Done once, modified as needed
3. Design - Once per feature
4. Plan - Once per feature
5. Milestone - One each week (ideally)
6. Incremental Prototypes - One to two weeks

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Roles

1. Project Manager - Ensures the team remains on track with completing both RIT deliverables and planned milestones.
2. Communicator - Coordinates all communications between the team and BAE sponsors. Additionally, they will be creating and sending out the weekly meeting agenda to BAE.
3. Hardware Manager - Manages equipment inventory, ensuring signout sheets are up to date and accurate, and doing periodic inventorying.
4. Methodology Overseer - Ensures the team is following established processes and will be responsible for managing the Jira board.

Standards and Quality Practices

1. Flight tests will be done regularly when appropriate.
2. There will be one milestone each week.
3. Code conventions will be established and followed.
4. Pull requests must be reviewed by at least one other person before merging to main.
5. Code that does not build or pass existing acceptance tests will not be merged to main.

Tools

1. Jira - Story tracking
2. Google Sheets - Timetracking
3. Powerpoint - Monthly Reports
4. GitLab - Source Control
5. AppGate - BAE Virtual Private Network

Metrics and Measurements

Performance Metrics

1. Time to Process and Report Detections
    a. Time in seconds from a dog entering the frame to notification appearing in the AR interface.
2. Machine Learning Metrics
    a. Recall
        i. Measures the rate of true positives compared to all positive instances in a dataset.
    b. Precision
        i. Measures the rate of true positives compared against the rate of false positives in a dataset.
    c. F1 Score

        i. Measures the harmonic mean between recall and precision, giving a general overview of model performance.
    d. mAP@0.5
        i. Measures the ability of a model to find detections and properly identify where those detections are in an image.
    e. Accuracy
        i. Measures the number of detections made against all available detections during test flights.

Per Plan

1. Time to completion
    a. The time in minutes for the drone to complete flying a plan
2. Total captured frames
    a. The number of frames captured by the drone while following a flight plan
3. Laps per battery (20 minute battery life)
    a. The number of laps the drone can complete when flying a flight plan before the battery dies
4. Distance Covered
    a. The distance in feet that the drone travels for a given flight plan

Project Metrics

1. Code Coverage
    a. The percentage of lines of code in newly created code covered by unit tests
2. Defect Escape Rate
    a. Number of reported defects in the code that has already been tagged as a release.

Process Metrics

1. Feature Slippage
    a. The ratio of the number of stories completed after the initially assigned date to the number of stories completed before or by the assigned date.
2. Percentage of Feature Completion
    a. The number of story points completed in an epic divided by the total number of points in that epic
3. Percentage of Project Completion
    a. The number of story points completed across all epics divided by the total number of points in all epics

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Planned Milestones and Major Deliverables

Epic: TigerRescue Android App

Milestones

- Get previous app compiling and running
- Confirm drone connection
- Confirm the drone can send & receive flight commands
- Fix drone autoflight code
- Create image pipeline to send geotagged images to mission control
- Send drone GPS data routinely

Epic: Sandtable Data Preparation

Milestones

- Load DEM
- Latitude/Longitude to Unity coordinate conversion
- Detection object creation

Epic: Sandtable AR Display

Milestones

- Display static object
- Display dynamic object
- Mesh agents for drones
- Mesh-attached callouts for dogs
- Colorcoding of explored/unexplored areas

Epic: AR Pop-Ups

Milestones

- Design unobstructive pop-up including available user actions
- Update map with image location (pin)
- Sync map pin with pop-up
- Pop-up queue

Epic: AR Input

Milestones

- Exploring AR input methods
- Prototyping input methods

- Accept/reject dog detection
- Gimbal control override
- Move to target location

Epic: Machine Learning Model

Milestones

- Detect any objects consistently
- Gather data for training on dogs
- Detect dogs consistently
- Find coordinates of dog detections
- Track persistent dogs between frames

Initial Requirements

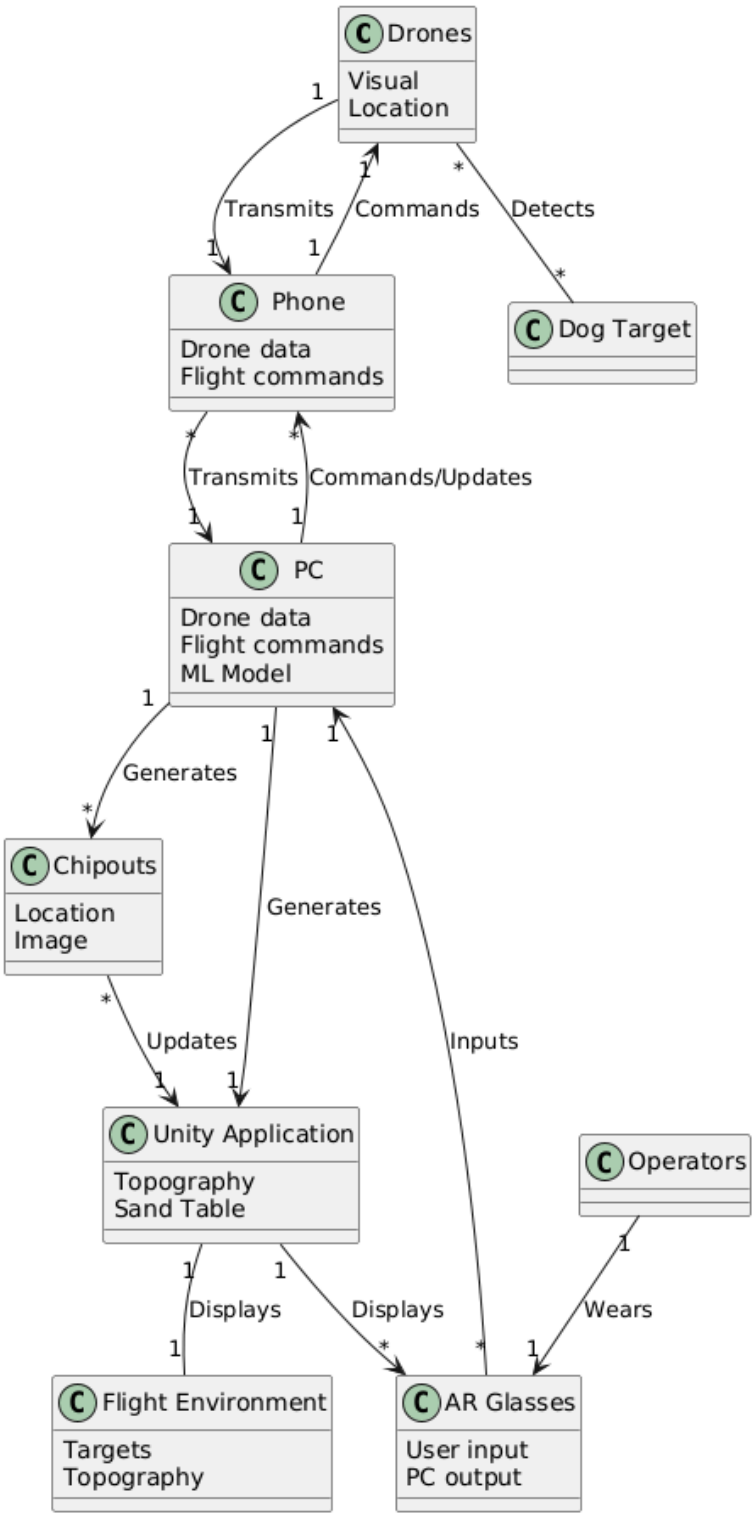
1. AR Sandtable
    a. The sandtable shall display local topography
    b. The sandtable shall show detections on the map
    c. The sandtable shall notify AR users of new detections
    d. The sandtable shall be synchronized for all AR users
    e. The sandtable shall visually display the the age of detection data
2. AR Synchronization
    a. The AR glasses shall display the sandtable with the same data at the same time
    b. The AR glasses shall display the same popups at the same time
    c. The AR glasses shall display the same detection pins at the same time
3. Callouts/Popouts
    a. The popups shall include location data of the detection
    b. The popups shall include chip-out images of detections
    c. The popups shall allow for user verification
4. Automated Dog Detection
    a. The model shall take in frames of imagery from the drones
    b. The model shall detect dogs in imagery frames
    c. The model shall draw bounding boxes around detected targets
    d. The model shall report confidence scores for each detection
    e. The model shall crop imagery so that a new chipout is generated for each detection
5. Autonomous Drone Flight
    a. The drones shall fly in a predetermined flight pattern
    b. The drones shall report data such as video and location to the onsite laptop
    c. The drones shall have a manual override to allow for user control

Communication and Stakeholder Management Plan

ID	Stakeholder Name	Role/Title	Interest in Project	Power	Communication Requirements
S1	Daniel Wallach & Michael Padovani	BAE Systems Project Sponsor	Customer, Advising	High	Weekly status meetings (agenda, 4up chart), monthly reports (project status, risk table), semester demos
S2	BAE	BAE Systems Project Sponsor	Customer	High	Monthly reports, semester demos
S3	Team	Senior Project Team	Creators	High	Discord, in-person meetings
S4	Samuel Malachowsky	Senior Project Advisor	Advising	Medium	RIT artifacts, weekly status meetings
S5	Operators	AR/Drone Operators	Users	Medium	Prototype demos, instructions for usage, presentation materials
S6	Dog Owners	Beneficiary	Beneficiary	Low	End of semester videos, end of year presentations and poster
S7	Dogs	Target	Beneficiary	Low	N/A

Risk Management Plan

Steps	Explanation	Artifacts
Identify knowns/unknowns	Analyze our known risks and identify any gaps that could potentially carry unforeseen risks.	4up, Weekly Meetings
Categorize risks	Break down risks into smaller categories based on what areas of the project they impact and who would need to deal with them.	Risk Table
Estimate impacts/delays	Project potential impacts/delays that could affect speed of delivery.	Risk Table
Decide if risk needs mitigation	Catalog which risks carry enough weight to warrant a mitigation plan. Risks without significant enough impact or downtime are excluded from our mitigation plan.	Risk Table

Create mitigation plan	Create and document the strategy for which we are using to mitigate the impact of the risk. This should be a clear and concise set of instructions or guidelines to minimize delays to the project.	Risk Table, Mitigation Strategy Document

Domain Model